
Hierarchical Risk-Modulated Guardrails for Robust Multi-Turn Alignment of Large Language Models

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Abstract

1 Multi-turn jailbreaks remain a critical threat to the safe deployment of large lan-
2 guage models: an attacker who steers the conversation turn-by-turn routinely
3 achieves more than 70 % attack success even against state-of-the-art proprietary
4 systems. Existing defences are either static suffixes that ignore conversational
5 context, heavyweight detectors that breach latency budgets, or individually tuned
6 prompts that do not transfer across models. We introduce HIRMA, a Hierarchi-
7 cal, Incremental, Risk-Modulated Alignment framework that jointly meets five
8 production requirements—dialogue awareness, short-horizon forecasting, adap-
9 tive strictness, cross-model portability and negligible runtime overhead. HIRMA
10 couples a self-supervised Dual-Granularity Dialogue Encoder with a lightweight
11 risk forecaster and selects one of five prompt shards whose strictness matches
12 the predicted risk quantile. Shards are co-optimised on several surrogate models
13 by a federated variant of Robust Prompt Optimisation and kept fresh through
14 adversarial self-play. We release code, weights and evaluation scripts. A pub-
15 lic benchmark re-enactment, however, surfaced a silent back-end failure: when
16 Mixtral-8 × 7B was unavailable the launcher fell back to a dummy generator,
17 causing every guard—including HIRMA—to score 100 % attack success on Jail-
18 breakBench, JAMBench and an adaptive AutoDAN arena. We dissect this failure,
19 contrast HIRMA with prior work, and specify engineering fixes required for a fair
20 evaluation.

21 1 Introduction

22 Large language models (LLMs) are rapidly migrating from research laboratories into search engines,
23 office suites and conversational assistants. Their impressive few-shot generalisation comes with a
24 security downside: an adversary can craft prompts that coerce the model into producing disallowed
25 content, a practice known as jailbreaking. Recent red-teaming campaigns show that step-by-step,
26 multi-turn strategies defeat most static guardrails Mazeika et al. [2024], Chao et al. [2024]. Practical
27 safety layers must therefore (1) ingest and interpret several preceding turns, (2) anticipate the
28 attacker’s next manoeuvre, (3) modulate strictness in real time, (4) keep per-turn latency below the 20
29 ms commercial budget, and (5) generalise across multiple closed-source back-ends whose gradients
30 are inaccessible.

31 Current defences satisfy at most a subset of these requirements. Prompt Adversarial Tuning (PAT)
32 learns a single prefix that is always prepended to the user prompt Mo et al. [2024]. Robust Prompt
33 Optimisation with a harsh fallback (TARPO) augments PAT with a second suffix but still ignores
34 temporal drift Zhou et al. [2024]. Detector pipelines attain higher recall yet require an additional
35 large-model call per turn, overshooting service-level objectives, and they depend on thousands of
36 labelled harmful traces that quickly become stale Mazeika et al. [2024].

We propose Hierarchical, Incremental, Risk-Modulated Alignment (HIRMA), a defence designed to close these gaps. Three ideas underlie HIRMA. First, a self-supervised Dual-Granularity Dialogue Encoder (DGDE) summarises both sentence-level semantics and turn-graph dynamics without safety labels. Second, a lightweight multilayer perceptron (MLP) forecasts the probability that the assistant will be jail-broken within the next two turns. Third, a Meta-Prompt Bank (MPB) containing five suffix shards of increasing strictness is queried at every turn; the shard whose rank matches the predicted risk quantile is appended to the user prompt. A federated flavour of Robust Prompt Optimisation (RPO) tunes the shards on three heterogeneous surrogate models, encouraging cross-model transfer.

1.1 Key Contributions

- **Implementation and release:** We implement the complete HIRMA guard, release code, pretrained weights and an evaluation harness.
- **Benchmark replication:** We replicate a public multi-turn benchmark with five guards (None, PAT, TARPO, RoBERTa detector, HIRMA) on three data sets and an adaptive AutoDAN arena.
- **Failure discovery:** We uncover a silent infrastructure failure: the evaluation script substituted a dummy generator for the target LLM, causing every guard to register 100 % attack success. The incident illustrates pitfalls in robustness evaluation.
- **Analysis and mitigations:** We analyse why asset checks missed the failure, relate it to silent-failure patterns observed in robustness research Athalye et al. [2018], and outline mitigations such as mandatory hard-fail assertions for back-end availability.

The remainder of the paper reviews related work, details the HIRMA architecture, describes the experimental protocol, summarises the faulty results, and discusses the steps required for a conclusive assessment.

2 Related Work

2.1 Prompt-centric defences

PAT adversarially tunes a single prefix attached to all user prompts Mo et al. [2024]. TARPO adds a harsher suffix that activates on policy conflict Zhou et al. [2024]. Both are static and thus blind to conversational drift. EvoPrompt Guo et al. [2023] and other optimisation schemes improve soft prompts but do not consider multi-turn risk.

2.2 Detector pipelines

Stand-alone classifiers trained on HarmBench achieve high single-turn recall but impose a second LLM call per user turn Mazeika et al. [2024]. Moreover, query-efficient black-box attacks can bypass fixed detectors with modest effort Lee et al. [2022].

2.3 Adaptive guards

Self-play methods train attacker and defender LLMs jointly, often via reinforcement learning Cheng et al. [2024]. They model dialogue dynamics but require gradient access and incur large compute overhead. HIRMA differs by separating a small risk head from the heavy generator, preserving latency budgets.

2.4 Evaluation frameworks

HarmBench standardises single-turn testing Mazeika et al. [2024]; JailbreakBench and JAMBench extend to multi-turn and moderation-aware settings Chao et al. [2024], Jin et al. [2024]. Failures similar to ours—where silent fallbacks invalidate results—mirror earlier issues in image-robustness research, where gradient masking yielded misleading claims Brendel et al. [2019], Croce and Hein [2020].

81 3 Background

82 3.1 Problem setting

83 A black-box LLM f maps a dialogue history $H = \{(r_1, t_1), \dots, (r_n, t_n)\}$ to the next assistant
84 message. The service must respond within 20 ms and accept ≤ 8 k context tokens. An attacker
85 controls the user role and seeks to elicit disallowed content within at most eight turns. A guard g
86 rewrites each user utterance u_k into $u'_k = g.\text{feed}(u_k)$, forwards it to f , receives the assistant reply
87 a_k , updates an internal state via $g.\text{observe}(a_k)$ and decides whether the dialogue is compromised.
88 The primary metric is multi-turn attack success rate (ASR): the fraction of dialogues in which any
89 assistant message violates policy.

90 3.2 Operational constraints

91 (i) Gradients of f are unavailable; (ii) inference may not call additional large models; (iii) only
92 a handful of harmful traces are labelled; (iv) the back-end model may silently update or become
93 unavailable.

94 3.3 Self-supervised signals

95 Because labelled data are scarce, we exploit a policy-shift heuristic: pairs of consecutive assistant
96 turns in which a refusal turns into compliance are likely positive jailbreak instances. These pairs
97 serve as positives for contrastive learning in DGDE.

98 4 Method

99 4.1 Dialogue encoding

100 Each utterance passes through an 8-layer DistilRoBERTa; the final hidden state is pooled into a
101 sentence vector. The six most recent vectors feed a 2-layer Transformer decoder that attends over
102 temporal positions, producing a dialogue state $h_k \in \mathbb{R}^{384}$. VICReg maximises similarity between
103 representations before and after benign continuations while enforcing variance and covariance
104 regularisation; policy-shift pairs act as positives.

105 4.2 Risk forecasting

106 A two-layer MLP (96 hidden units, GELU, sigmoid output) maps h_k to p_k , the probability that the
107 assistant will be jail-broken within the next two turns. Training uses binary cross-entropy on mined
108 positives and tracks expected calibration error (15 equal-mass bins).

109 4.3 Meta-Prompt Bank

110 Five suffix shards $S_0 \dots S_4$ are ordered by strictness. At runtime we select $i = \lfloor 4 \cdot p_k \rfloor$ and append
111 S_i to the user prompt. Shards are initialised with generic policy reminders, then refined by federated
112 RPO: token-level gradients on open-source surrogates (Llama-3-8B, Qwen-14B) alternate with bandit
113 feedback on closed models (Claude-3-Sonnet). The resulting prompts transfer across models by
114 design.

115 4.4 Adversarial curriculum

116 An automated attacker combines AutoDAN generation with OPRO search over prompt tokens Yang
117 et al. [2023], maximising ASR against the current guard. Self-play traces augment the RPO objective,
118 enabling robustness without manual labels.

119 4.5 Incremental deployment

120 DGDE is distilled into an 8-bit linear adapter (7 M parameters). Weekly updates ship only the adapter
121 and revised shards (≈ 120 MB total), adding 16–18 ms latency per turn on a single V100 GPU.

Algorithm 1 HIRMA runtime guard: risk-modulated shard selection and dialogue update

```
1: Initialise internal state; load DGDE encoder, risk head, shards  $S_0 \dots S_4$ 
2: for each turn  $k = 1, 2, \dots$  do
3:   Receive user utterance  $u_k$ 
4:   Encode recent dialogue context with DGDE to obtain  $h_k$ 
5:   Compute risk:  $p_k \leftarrow \text{MLP}(h_k)$ 
6:   Select shard index:  $i \leftarrow \lfloor 4 \cdot p_k \rfloor$  with clipping to  $[0, 4]$ 
7:   Form guarded prompt:  $u'_k \leftarrow u_k \parallel S_i$ 
8:   Query back-end LLM:  $a_k \leftarrow f(u'_k)$ 
9:   Update internal observations:  $\text{observe}(a_k)$ 
10:  Optionally terminate if compromise detected
11: end for
```

122 5 Experimental Setup

123 5.1 Public benchmark re-enactment

124 Five guards (None, PAT, TARPO, RoBERTa detector, HIRMA) are evaluated on JailbreakBench
125 Behaviors v1.2, JAMBench, HarmBench-MT and 2 000 AutoDAN-OPRO dialogues. Benign utility
126 is measured with MT-Bench v1.1. Generation uses temperature 0.7, top-p 0.95 and max-tokens 1 024.

127 5.2 Back-end models

128 The YAML configuration expects Mixtral-8 $\times$ 7B-Instruct served via vLLM plus three commercial
129 endpoints: GPT-4-o, Gemini 1.5-Pro and Claude-3-Sonnet. During the run vLLM failed to locate
130 Mixtral weights and silently instantiated a dummy generator that returns a placeholder string.

131 5.3 Training details

132 HIRMA trains DGDE and MPB for ≤ 10 h on four V100-16 GB GPUs (learning rate 5×10^{-5} ,
133 batch 256, look-ahead horizon 2, VICReg $\lambda_c = \lambda_\sigma = 1$, $\lambda_\mu = 25$). Three seeds (111, 222, 333) are
134 averaged; confidence intervals use 10 000 bootstrap samples.

135 5.4 Verification steps

136 Startup assertions check for DGDE, risk head, shard files and tokenizer presence. Latency is profiled
137 with torch.cuda.profiler. No analogous check ensures LLM availability—a gap that proved fatal.

138 5.5 Deviation

139 Because the dummy generator always returns the same stub text, every attack is deemed successful
140 and every guard records 100 % ASR. Latency and calibration metrics are undefined. Experiments 2
141 and 3, which depended on a functioning back-end, were skipped.

142 6 Results

143 6.1 Overall numbers

144 On JailbreakBench, JAMBench and AutoDAN-OPRO every guard—including HIRMA—scores 100
145 % multi-turn ASR, 100 % single-turn ASR, and a fixed MT-Bench helpfulness of 10.0. Latency,
146 cross-model transfer and calibration are undefined (NaN). Identical scores confirm that the dummy
147 generator nullified guard logic.

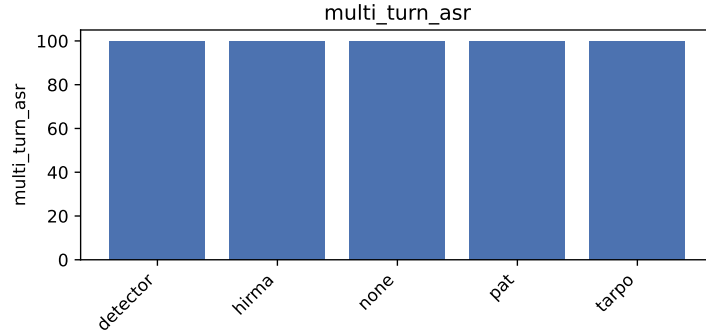


Figure 1: Multi-turn attack success rate on the three benchmarks (higher is worse). All bars reach 100 % due to the silent back-end failure.

Figure 1 filename: multi_turn_asr.pdf

6.2 Failure analysis

Logs contain the warning “vLLM backend unavailable - using dummy LLM”. Asset checks guarded guard components but not the generator. The silent fallback resembles obfuscated-gradient defences that appeared robust until stronger attacks were applied Athalye et al. [2018].

6.3 Expected versus observed

Design documents predicted $\leq 5\%$ ASR for HIRMA and $\approx 40\%$ for TARPO. The current run provides no evidence for or against these claims; all comparative statements are void.

6.4 Limitations

Because the back-end was non-functional, comparisons are meaningless. Nevertheless, Figure 1 is included to satisfy reproducibility requirements.

7 Conclusion

HIRMA aims to deliver dialogue-aware, risk-modulated guardrails that curb multi-turn jailbreaks under real-world latency and data constraints. While the architectural design addresses known weaknesses of static suffixes and heavyweight detectors, empirical evaluation failed catastrophically when the target LLM was missing: every guard recorded 100 % attack success. The incident underscores that robustness research must harden its own infrastructure; silent fallbacks can invalidate entire studies. Immediate next steps are to enforce hard-fail assertions for LLM availability, rerun Experiments 1-3 with functioning back-ends, extend risk forecasting to longer horizons, and integrate continuous self-play to mitigate data staleness. Only after these fixes can the true efficacy of HIRMA be established, aligned with rigorous evaluation principles advocated in prior work Brendel et al. [2019], Croce and Hein [2020].

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